

SET 10

1. R: Create a numeric vector with missing values; identify, count and replace missing values using the mean.

The screenshot shows the RStudio interface with the following content:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
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Platform: x86_64-w64-mingw32/x64

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'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> # Create a numeric vector with missing values
> x <- c(10, 20, NA, 30, 40, NA, 50, 60)
>
> # Identify missing values
> is_missing <- is.na(x)
> print(is_missing)
[1] FALSE FALSE TRUE FALSE FALSE TRUE FALSE FALSE
>
> # Count missing values
> missing_count <- sum(is.na(x))
> cat("Number of missing values:", missing_count, "\n")
Number of missing values: 2
>
> # Calculate mean excluding missing values
> mean_value <- mean(x, na.rm = TRUE)
> cat("Mean value:", mean_value, "\n")
Mean value: 35
>
> # Replace missing values with the mean
> x[is.na(x)] <- mean_value
>
> # Display the final vector
> cat("Vector after replacing missing values:\n")
Vector after replacing missing values:
> print(x)
[1] 10 20 35 30 40 35 50 60
>
```

The Environment pane on the right shows the following variables:

Variable	Value
is_missing	logi [1:8] FALSE FALSE TRUE FALSE FALSE TRUE ...
j	3
MAD	6.16326530612245
mean_value	35
mean_x	24.1428571428571
missing_count	2L
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
x	num [1:8] 10 20 35 30 40 35 50 60

2. R: Create 10 employee records and sort by Salary descending; display the top three.

The screenshot shows the RStudio interface with the following content:

```
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Type 'q()' to quit R.

[Workspace loaded from ~/.RData]

> # Create 10 employee records
> employees <- data.frame(
+   Name = c("Arun", "Bala", "Chitra", "Divya", "Esha",
+           "Farhan", "Goku", "Hari", "Isha", "Jaya"),
+   Age = c(25, 30, 28, 35, 26, 32, 29, 40, 27, 31),
+   Salary = c(35000, 52000, 45000, 70000, 40000,
+            60000, 48000, 80000, 42000, 55000)
+ )
>
> # Sort employees by Salary in descending order
> employees_sorted <- employees[order(-employees$Salary), ]
>
> # Display top 3 employees
> top_three <- head(employees_sorted, 3)
>
> print(top_three)
   Name Age Salary
8  Hari  40 80000
4 Divya  35 70000
6 Farhan 32 60000
>
```

The Environment pane on the right shows the following variables:

Variable	Value
employees	10 obs. of 3 variables
employees_sorted	10 obs. of 3 variables
top_three	3 obs. of 3 variables

The Values pane shows the following data for the selected variable:

Variable	Value
j	3
MAD	6.16326530612245
mean_x	24.1428571428571
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
x	num [1:7] 14 18 20 23 27 31 36

3. WEKA: Load a dataset with missing values and use ReplaceMissingValues; compare before and after preprocessing.

Dataset: Missing Values

```
@relation employee_missing
@attribute Age numeric
@attribute Salary numeric
@attribute Department {IT,HR,Sales}
@data
25,35000,IT
30,42000,HR
,48000,IT
35,,Sales
40,60000,HR
28,39000,Sales
,52000,IT
45,70000,HR
```

The screenshot shows the Weka Explorer application with the 'Preprocess' tab selected. The 'ReplaceMissingValues' filter is chosen in the 'Filter' dropdown. The 'Current relation' is 'employee_missing-weka.filters.unsupervised.attribute.Repl...'. The 'Attributes' list shows 'Age', 'Salary', and 'Department'. The 'Selected attribute' section shows 'Age' with 'Missing: 0 (0%)' and 'Distinct: 7'. The 'Class' is set to 'Department (Nom)'. A bar chart is displayed below the class selection, showing the distribution of the 'Department' attribute. The chart has three bars: a cyan bar for 'IT' (value 6), a red bar for 'HR' (value 2), and a blue bar for 'Sales' (value 2). The x-axis is labeled with values 25, 35, and 45. The status bar at the bottom shows 'OK' and a 'Log' button.

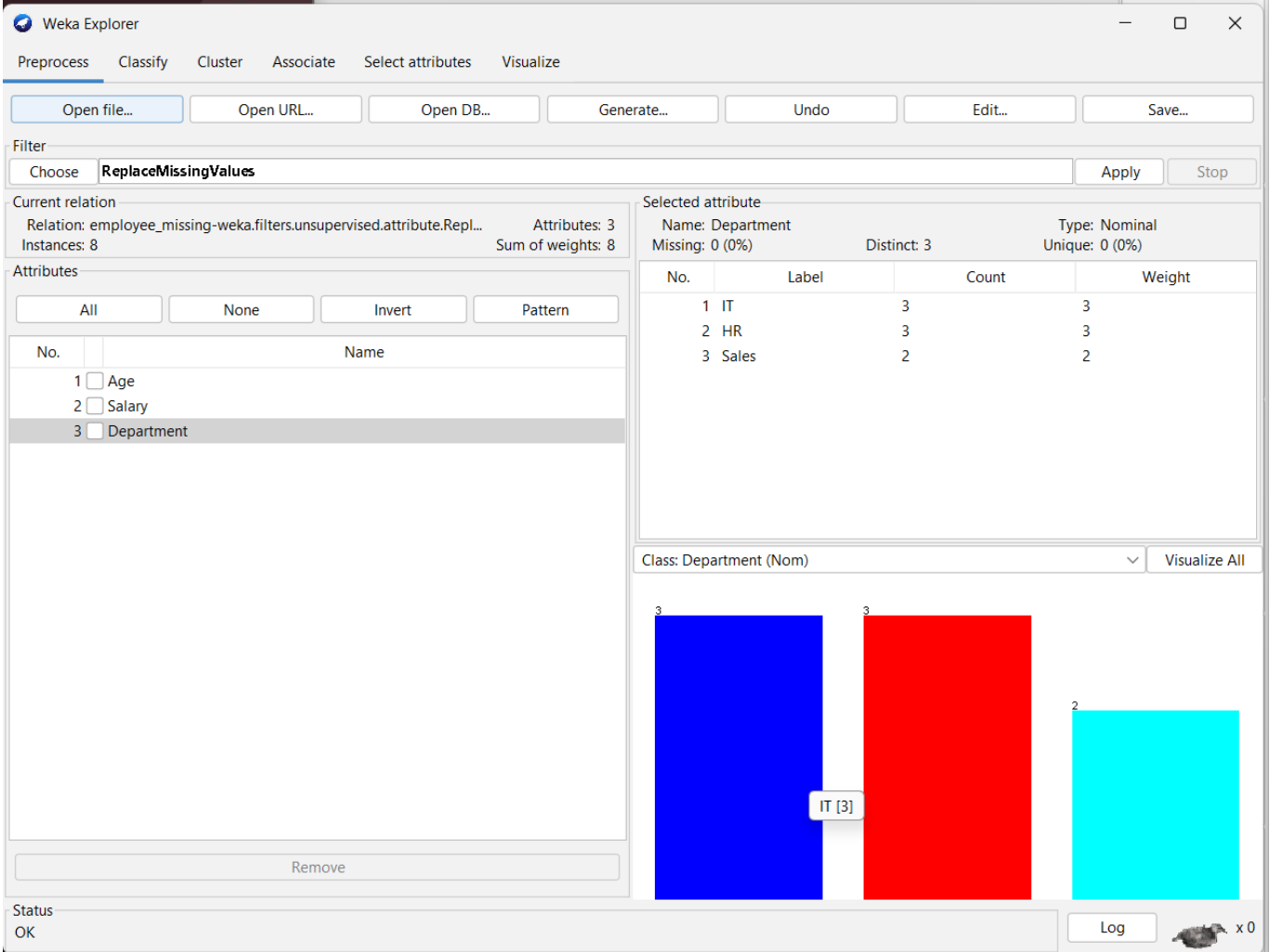
Statistic	Value
Minimum	25
Maximum	45
Mean	33.833
StdDev	6.446

Department	Count
IT	6
HR	2
Sales	2

4. WEKA: Remove one irrelevant attribute with the Remove filter and compare classifier performance before and after.

Dataset: Attribute Removal

Use the same Employee Missing dataset from Set 10 Q3. Remove Department as the irrelevant attribute after handling missing values.



The screenshot shows the Weka Explorer application window. The 'Preprocess' tab is active. The 'Filter' dropdown is set to 'ReplaceMissingValues'. The 'Current relation' is 'employee_missing-weka.filters.unsupervised.attribute.Repl...'. The 'Attributes' list shows 'Age', 'Salary', and 'Department'. The 'Selected attribute' section shows 'Department' with 'Missing: 0 (0%)' and 'Distinct: 3'. A bar chart visualizes the distribution of the 'Department' attribute with three categories: IT (3), HR (3), and Sales (2).

No.	Label	Count	Weight
1	IT	3	3
2	HR	3	3
3	Sales	2	2

Class: Department (Nom) Visualize All

Status: OK